

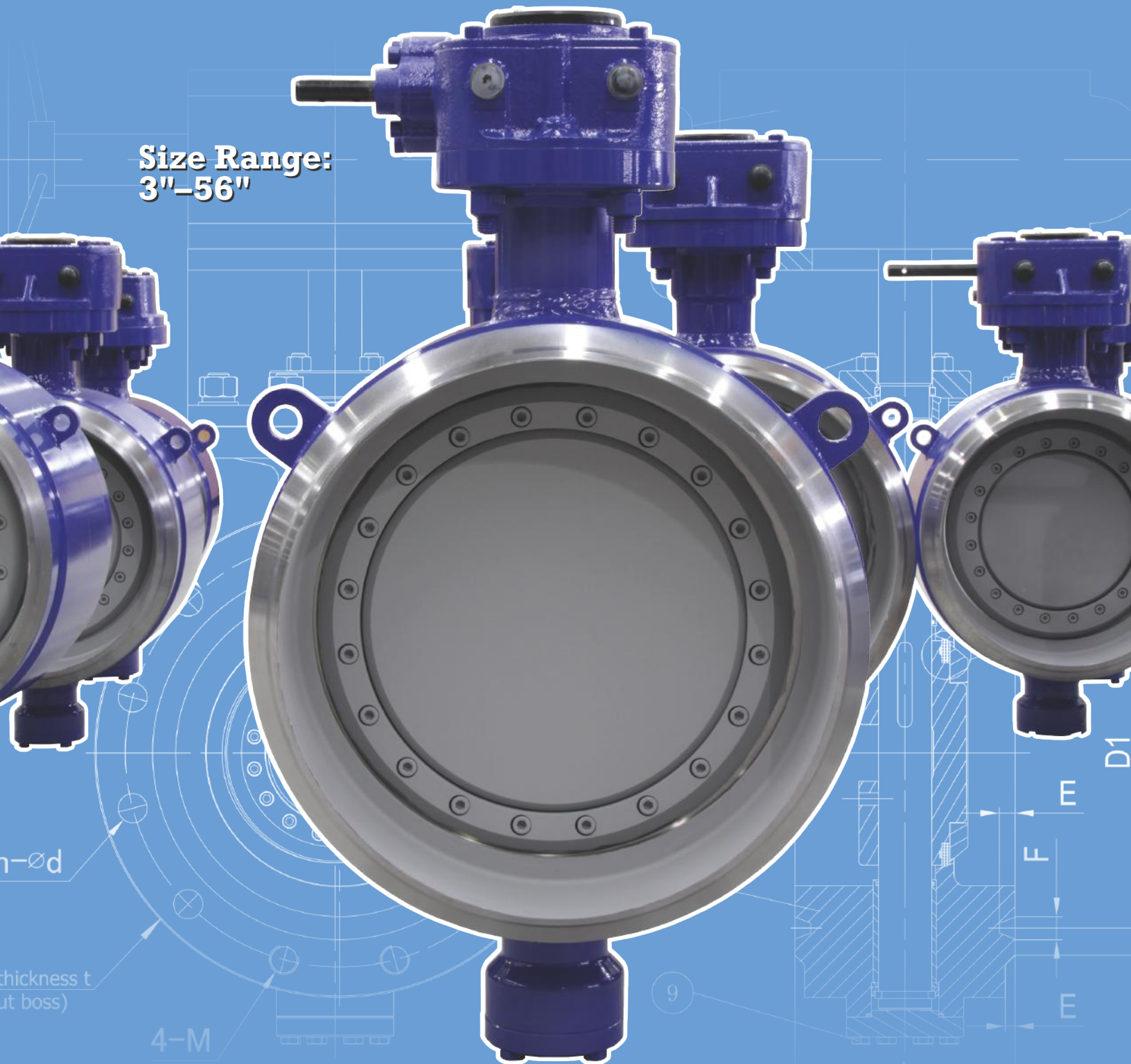


Triple Offset High Performance

Z-Tek Series

Engineered Valves for Severe Service Reliability with Innovative Solutions.

Size Range:
3"–56"



Demanding services requiring resistance to abrasion, erosion, corrosion, solids, and high temperatures beyond the capabilities of most butterfly valves.

Butt-Weld — Flanged — Wafer — Lug



Triple Offset Butterfly Valve Z-TEK Series

Product Features

- The Z-TEK valve utilizes a triple offset structure and has no friction with seat during the process of 90° full travel operation.
- Torque sealing instead of position sealing is adopted to achieve bi-directional sealing with long-term stability. It can meet zero leakage in the positive flow direction.
- All-metal sealing structure is adopted and the requirements on leakage class are higher than those of **API 6D** and **API 598** on metal sealed valves.
- The valve sealing surface overlay is stellite. The valves can be used in abrasive working conditions and ensures reliability. Meanwhile, there is no need to replace the body. This ensures low maintenance cost.
- Integrated casting body and the face-to-face dimensions conforms to the requirements of **ISO 5752**, **ASME B16.10** and **API 609**. Compared with other valves, the Z-TEK is assembled with a superior, flexible design.
- The all-metal construction and sealing components meet the requirements of fire safe standards. The valve complies with **API 607**.
- The triple offset structure design allows excellent flow capacity and excellent regulating abilities.



General Application

- | | |
|---|--------------------------------|
| • Petroleum and natural gas processing | • Chemicals and chemical plant |
| • Water treatment plant and supply system | • LNG |
| • Offshore drilling platform | • Storage and Logistic |
| • Desalination plant | • Power plant |
| • Oil refining | • Papermaking |
| • District heating | • Steel plant |



Technical Data

Size Range:

3"~56" (DN80~1400)

Pressure rating:

ASME Class 150~900

Temperature:

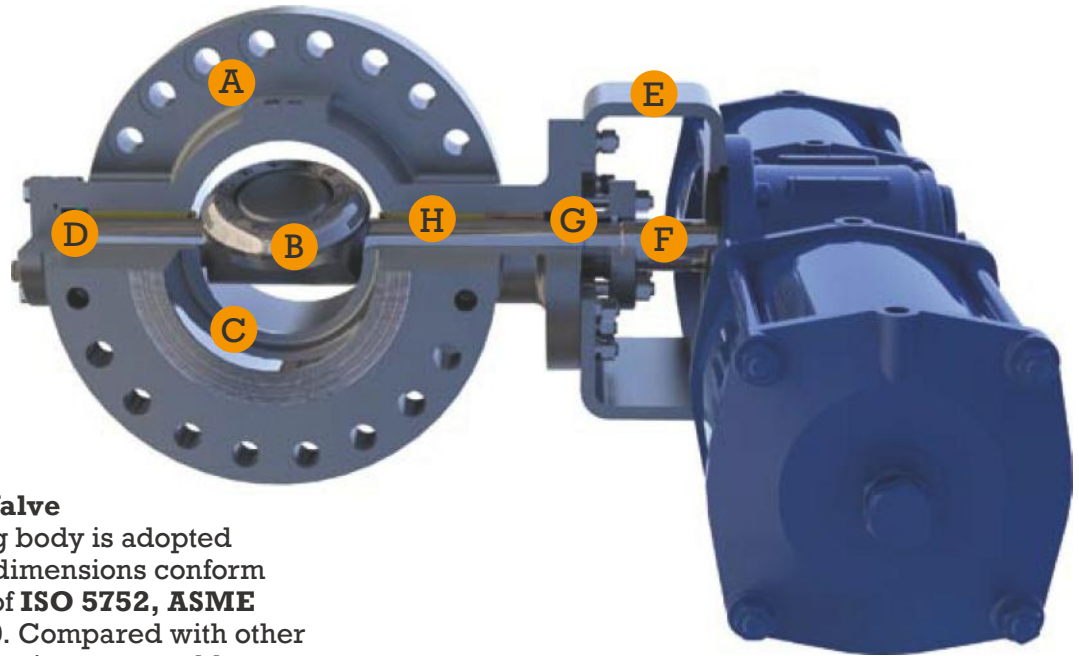
-76°F (-60°C)~842°F (450°C)

Cryo Temperature Range:

-340°F (-206°C)~100°F (38°C)

Triple Offset Butterfly Valve Z-TEK Series

Design Characters



A Integrate Casting Valve

An integrated casting body is adopted and the face-to-face dimensions conform to the requirements of **ISO 5752, ASME B16.10, and API 609**. Compared with other valves, this valve is easier to assemble.

B Flexible Metallic Sealing Ring

A flexible metallic sealing ring is installed on disc. A retainer ring compresses the sealing ring onto the valve plate, which can ensure the fixture of the sealing ring and guarantee that there is some flexible deformation. The sealing ring is positioned through the pin, making the sealing ring replacement easier. The retainer ring with thickening design is equipped with multiple socket head screw compression sealing rings, which ensures the hardness of the whole valve plate is more reliable.

C Metallic Sealing Seat

The valve sealing surface overlay stellite ensures the durability of the valve in various applications including under the working conditions such as high temperature and sanitation. The stellite alloy surfaced seat is a standard configuration for all specifications.

D High Cycle Thrust Bearing

The high cycle thrust bearing presses a high hardness thrust bearing onto the shaft and forms a central sliding gasket extending its service life.

E Top Flange

The top flange adopts **ISO 5211** standard and can be directly connected to various actuators.

F Blow-Out Prevention

The standard external blow-out prevention design can prevent the valve stem from blowing out under pressure and improve the safety of the valve.

G Valve Stem Packing Seal

The valve stem packing seal may adopt multiple structural forms in a flexible way, according to different working conditions of the valve media. The packing design could prevent fugitive emissions and such sealing conforms to **ISO 15848:1 Class BH** and **verification of API 622**, including washing equipment for packing, bearings, and testing device for packing. Measurement for fugitive emissions conforms to **EPA and TA Lust standards**.

H Valve Stem Bearing

Long bearings adopt surface hardening treatment, which can reduce the friction of the valve stem and reduce torque. Applicable to the working conditions of unclean media or polymer, the bearing retainer can provide better protection for the bearings and extend their service life.

Max-Seal offers a broad line of automation systems for precise proportioning or off-off control in either pneumatic or electrically powered units.

Triple Offset Butterfly Valve Z-TEK Series

Exploded View

Triple Offset Eccentric Valve Model Number ID Codes

Model	Pressure Class	Body		Disc		Stem		Seat		Seal Ring		Operator		Size in	Size mm
Triple Wafer-BWTZ Flanged-BFTZ Lug-BLTZ	150	316 SS	SS	316 SS	SS	17-4PH	S7	STELLITE GR.21	ST	316SS + Graphite	SG	Gear	G	3"	80
	300	WCB	CS	WCB	CS	410SS	S1	13% Chrome	CR	Inconel 625 + Graphite	IG	Bare Stem	N	4"	100
	600	WC6	W6	WC6	W6	XM-19	XM			316SS Solid Metal	SM	Actuator	A	6"	150
	900													8"	200

(Always add "T" for triple offset)

Size Range: 3"-56"

Ordering Example

Model	Pressure Class	Body		Disc		Stem	
Flanged	150	316SS	SS	316SS	SS	17-4PH	S7
BFT	600	CS		SS		S7	

Seat		Seal Ring		Operator		Size mm
STELLITE	ST	316SS Solid Metal	SM	Gear	G	80
ST		SM		G		80

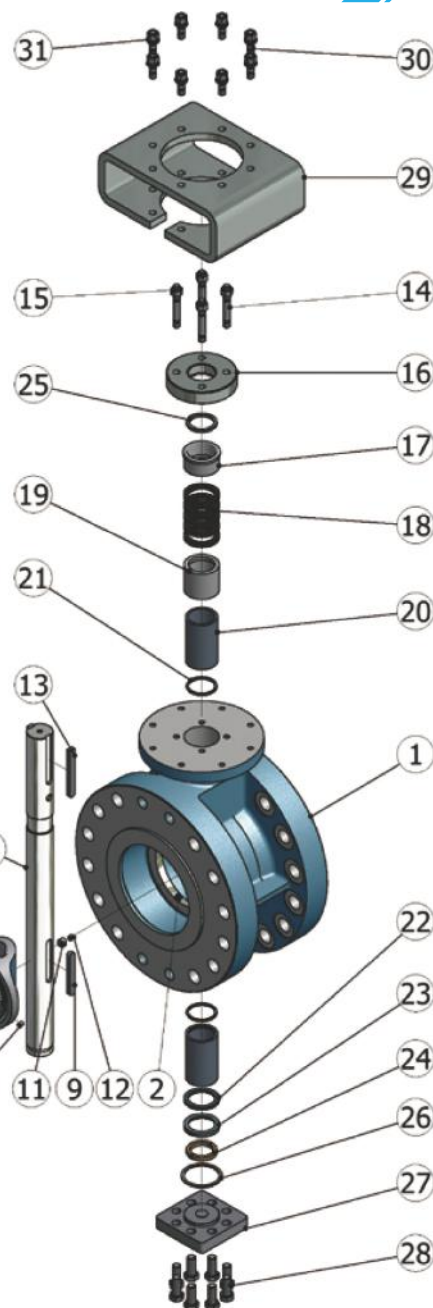
Design Advantage

- The triple offset butterfly valve has been designed to answer the industries demand for an alternate solution to gate valves and the ability to modulate flow.
- Lower weight vs Gate or Globe valve.
- Zero leakage shut off to API 598.
- Easy to automate.
- Ideal for modulating for process control.
- Fire Safe Certified to API 607-4



Engineered Valves incorporate the most innovative designs on today's market. They provide uninterrupted service in severe applications.

Exploded View Parts List



Triple Offset Butterfly Valve Z-TEK Series

Standard Bill of Material

Item	Part	Carbon Steel	Stainless Steel	Alloy Steel
1	Body	ASTM A216 WCB	ASTM A351 CF8M	ASTM A217 WC6
2	Seat	Stellite*	Stellite*	Stellite*
3	Disc	ASTM A216 WCB	ASTM A351 CF8M	ASTM A217 WC6
4	Stem	A479 410	A479 XM-19	A479 410
5	Spiral wound gasket	316SS + graphite	316SS + graphite	316SS + graphite
6	Seal ring	A276 S31803	A276 S20910	A276 S31803
7	Retainer ring	ASTM A105N + ENP	ASTM A182 F316	ASTM A182 F11
8	Bolt	A2-70	A2-70	A2-70
9	Key	ASTM A276 410	ASTM A182 F316	ASTM A276 410
10	Pin	ASTM A276 410	ASTM A182 F316	ASTM A276 410
11	Nut	ASTM A276 410	ASTM A182 F316	ASTM A276 410
12	Screw	Stainless steel	Stainless steel	Stainless steel
13	Key	AHSI 1045	AHSI 1045	AHSI 1045
14	Bolt	ASTM A193 B7M	ASTM A193 B8M	ASTM A193 B16
15	Nut	ASTM A194 2HM	ASTM A194 8M	ASTM A194 4
16	Flange gland	ASTM A105N	ASTM A182 F316	ASTM A182 F11
17	Flange gland	ASTM A276 410	ASTM A182 F316	ASTM A276 410
18	Packing	Graphite	Graphite	Graphite
19	Spacer	ASTM A276 410	ASTM A182 F316	ASTM A276 410
20	Bushing	ASTM A276 316	ASTM A276 316	ASTM A276 316
21	Dust ring	Graphite	Graphite	Graphite
22	Bearing washer	ASTM A276 316	ASTM A276 316	ASTM A276 316
23	Thrust bearing	ASTM A276 316	ASTM A276 316	ASTM A276 316
24	Anti-drop gasket/spacer collar	ASTM A276 410	ASTM A182 F316	ASTM A276 410
25	Spacer collar	ASTM A276 410	ASTM A182 F316	ASTM A276 410
26	Bottom wound gasket	316SS + graphite	316SS + graphite	316SS + graphite
27	Bottom flange	ASTM A105N	ASTM A182 F316	ASTM A182 F11
28	Bottom screw	ASTM A193 B7M	ASTM A193 B7M	ASTM A193 B16
29	York	Q235	Q235	Q235
30	Bolt	ASTM A193 B7M	ASTM A193 B8M	ASTM A193 B16
31	Screw	ASTM A194 2HM	ASTM A194 8M	ASTM A194 4

*Seat Available in Gr. 6, Gr. 12, Gr. 21

Notes: The standard supply material include WCB, CF8M, LCB, WC6, CF3M and duplex stainless steel. In addition, special materials may be supplied according to user demands. Z-TEK shall have the right to modify the above materials without further notice.

Standards and Specifications

Quality Warranty	- ISO 9001 - API Q1	Top Flange Boring Standard	ISO 5211 Industrial Valves Quarter-turn Actuator Attachments
Design Standard	- ASME E B16.34 Valves-Flanged, Threaded, and Welding End - API 609 Butterfly Valves: Double-flanged, Lug- and Wafer type	Testing Standard	- API 598 Valve Inspection and Testing - ISO 5208 Industrial Valves Pressure Testing of Metallic Valves
Face to face	- ASME B16.10 Face-to-Face and End-to-End Dimensions of Valves - ISO 5752 Metal valves for use in flanged pipe systems Face-to-Face and centre-to-face dimensions - API 609 Butterfly Valves: Double-flanged, Lug- and Wafer type	Flange Drilling Standard	- ASME B16.5 Pipe Flanges and Flanged Fittings - ASME B16.47 Large Diameter Steel Flanges - DIN 2501 - Pipe flanges part 1: steel flanges for industrial and general service piping systems - EN 1092
Fire Safe Testing	API 607, 4th Edition Fire Test for Soft Seated Quarter-Turn Valves	Marking Standard	MSS SP-25 Standard Marking System for Valves, Fittings, Flanges, and Unions

Triple Offset Butterfly Valve Z-TEK Series

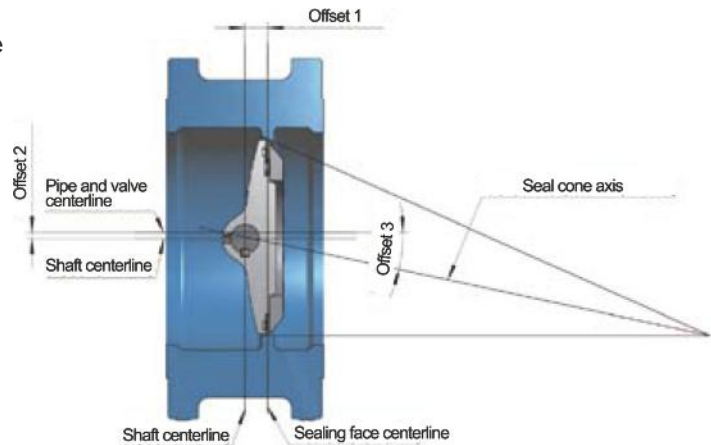
Design Characters of Sealing Structure

Triple Offset Structure

Offset 1: The shaft is offset from the sealing face centerline.

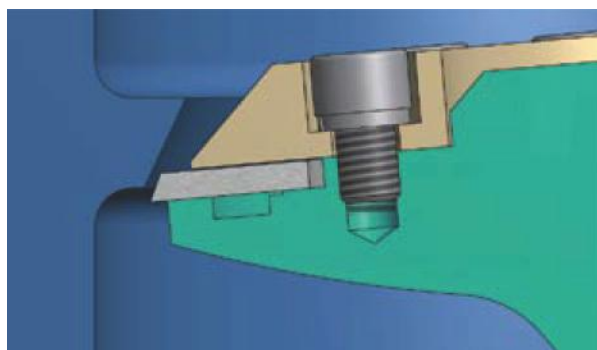
Offset 2: The shaft centerline is offset from the pipe and valve centerline. The two parts design above reduces the friction between the disc and seat.

Offset 3: The seat cone axis is offset from the shaft centerline to eliminate friction during closing & opening and to achieve uniform compressive sealing around the entire seat. The calculation and selection of contact angle for triple offset design ensures the valve will not be jammed. Meanwhile, the feature of torque seal ensures that the torque required by the valve switch keeps stable during the process of operation and guarantees a longer service life.

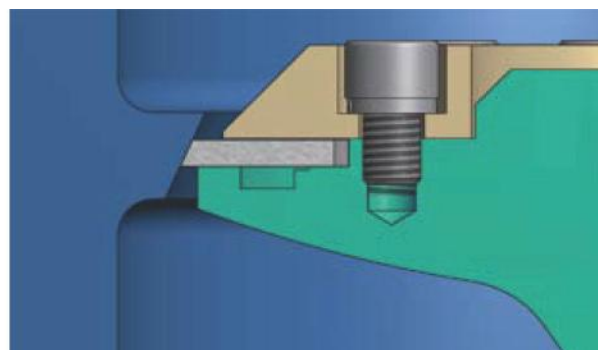


Flexible Metallic Valve Seat

The triple offset butterfly valve uses the flexible stainless steel installed on the disc to realize true zero leakage. *Zero leakage* is enabled through the flexibility of the sealing ring on the disc. When any torque is applied to the stem, the flexibility of the sealing ring is generated through the axial compression and flexible movement of the sealing ring. The contact face between the sealing ring and valve seat is “cone in cone” and the two cones are inclined. The contact face can reduce the slight “sledge effect” and make the sealing ring produce flexible movement and axial compression. Due to the even contact between the seat and sealing ring, as well as the flexible movement of the sealing face, even load is generated surrounding the valve seat so as to realize the tightest cutting using the smallest torque. The flexibility generated by the torque can make the valve shut off tight, regardless of flow direction or media pressure.



Prior to seating



Seated

Max-Seal's focus is Safety - Performance - Value

Triple Offset Butterfly Valve Z-TEK Series

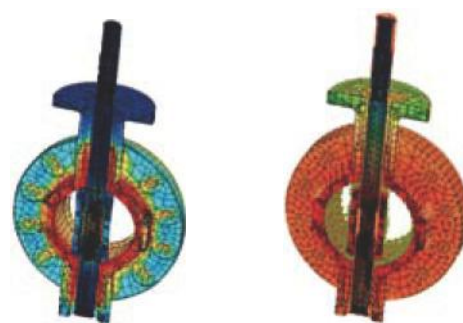
3D CAD/CAE Design

In order to design quality valves which conform to most standards and requirements and ensure their performance will fully satisfy the demands of most industries, the Z-TEK Series uses 3D solid for the elite design. Through the use of such method, the design structure of important parts can be analyzed, thus optimizing product structure design.

The Z-TEK body acts as the main bearing component of the valve; stem, and disc are key parts to pass on torque and produce the effect of sealing; sealing elements are the most core parts used to ensure the sealing performance of a valve. In order to ensure the safety and reliability of products, strength analysis under various working conditions are made using CAE software and tests the stress of the valve body and stem. Our designers exhibit 3D models as well as necessary analysis and calculation results to our design engineers. Such feedback from the design to the manufacturing process and then from manufacturing process to design, enables the improvements of product design and superior manufacturing quality.

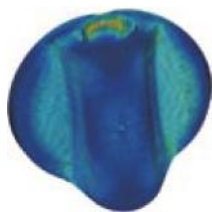
Thermal Stress Analysis

Under thermal load, the valve will produce mechanical deformation and stress. Through simulation of working conditions at high temperatures, we can correctly understand the stress distribution and changes under certain thermal loads, so as to optimize structural design and satisfy the operating performance of the valve.

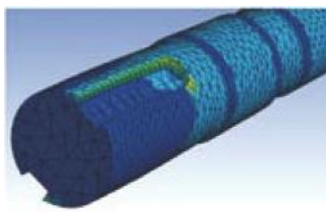


Stable Temperature Field
Distribution Chart

Stress Distribution Chart at
Stable Temperature Field



Disc Strength Analysis



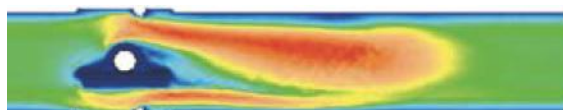
Stem Strength Analysis

Strength Analysis

Through simulation of working conditions on all the parts, we can correctly understand the stress distribution and changes of valve parts under certain thermal load, to optimize structural design and satisfy the operating performance of the valve.

Analysis on Flow Field Speed and Pressure in Pipelines.

Through simulation and analysis, we can understand the flow capacity of the Z-TEK Butterfly Valve Series, obtain the speed and pressure distribution of fluids, translate them into relevant Cv valves, and provide reference for later design and optimization.



Flow Field Speed Distribution Chart in Pipelines



Flow Field Pressure Distribution Chart in Pipelines

Triple Offset Butterfly Valve Z-TEK Series

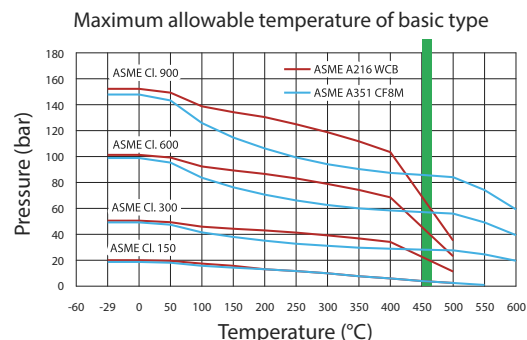
Technical Parameters

Temperature-Pressure Curve

Notes: The temperature-pressure grade of all valves fully conforms to the requirements of **ASME B16.34**

It is not recommended to use **ASTM A216 WCB** under the working conditions of temperatures higher than 797°F (420°C)

Torque Data (in-lbs) (Nm)

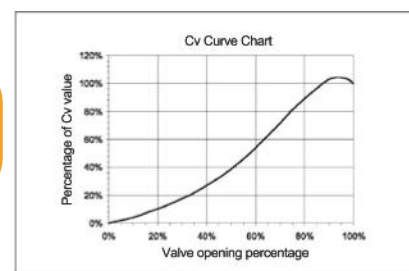


Size		Differential Pressure (bar)															
inch	mm	20				50				100				150			
		Closed		Open		Closed		Open		Closed		Open		Closed		Open	
3	80	700	79	788	89	1186	134	1363	154	1894	214	2868	324	7080	800	18762	2120
4	100	956	108	1169	132	1664	188	2089	236	3381	382	6364	719	21240	2400	43950	4966
6	150	1328	150	2363	267	2956	334	3806	430	6195	700	14966	1691	39825	4500	74296	8395
8	200	2266	256	3744	423	4647	525	9320	1053	14745	1666	27294	3084	62481	7060	108608	12272
10	250	4204	475	6638	750	8576	969	15258	1724	23701	2678	46844	5293	86853	9814	144326	16308
12	300	5983	676	9930	1122	12134	1371	22453	2537	30701	3469	68243	7711	119024	13449	193568	21872
14	350	7240	818	13532	1529	14647	1655	29055	3283	40073	4528	81544	9214	172098	19446	268474	30336
16	400	9541	1078	18577	2099	19347	2186	49676	5613	44144	4988	118635	13405	258642	29225	379674	42901
18	450	12134	1371	24630	2783	24515	2770	56685	6405	70128	7924	214011	24182	405286	45795	580525	65596
20	500	13187	1490	29002	3277	26550	3000	73694	8327	93456	10560	276820	31279	-	-	-	-
24	600	19134	2162	48286	5456	38268	4324	113466	12821	98802	11164	385214	43527	-	-	-	-
26	650	33223	3754	91970	10392	61446	6943	154212	17425	128989	14575	521912	58973	-	-	-	-
28	700	48304	5458	162743	18389	71473	8076	211409	23888	176877	19986	690823	78059	-	-	-	-
30	750	55393	6259	221003	25311	131954	14910	379701	42904	262633	29676	871725	98500	-	-	-	-
32	800	63889	7219	238065	26900	131954	14910	395525	44692	316653	35780	1069877	120890	-	-	-	-
36	900	94094	10632	321424	36319	218826	24726	617191	69739	377692	42677	1243248	140480	-	-	-	-
40	1000	116458	13159	382091	44689	270403	30554	710540	80287	-	-	-	-	-	-	-	-
42	1050	129308	14611	509079	57523	260155	29396	1116587	126168	-	-	-	-	-	-	-	-
48	1200	202250	22853	625173	70641	363603	41085	1318208	148950	-	-	-	-	-	-	-	-
56	1400	299927	33890	693690	78383	-	-	-	-	-	-	-	-	-	-	-	-

Notes: 1. The torque listed above is the values at normal temperatures and when the media flow is in the tangential direction of the valve stem.
2. The torque will change according to different media and internal parts.

Notes: 1. Definitions Cv: Flow of water when it passes through the valve at the temperature of 59°F (15°C) and under the pressure drop of 1 PSI (gal/min) Kv: Flow of water when it passes through valve at the temperature of 59°F (15°C) and under the pressure drop of 1 bar (m3/h)

2. The flow direction is in the tangential the direction of the valve stem. 3. $Cv=1.156Kv$



Valve Cv Values (Flow coefficient)

Size		150# Class		300# Class		600# Class		900# Class	
Inch	mm	Cv	Kv	Cv	Kv	Cv	Kv	Cv	Kv
3"	80	94	81	94	81	150	128	-	-
4"	100	210	180	210	180	250	214	-	-
6"	150	790	677	630	540	600	514	500	428
8"	200	1530	1310	1250	1071	1080	925	930	797
10"	250	2589	2217	2383	2041	1700	1456	1450	1242
12"	300	3923	3360	3579	3065	2520	2158	2150	1841
14"	350	5195	4450	4627	3963	4068	3484	3140	2689
16"	400	6940	5944	6251	5354	5380	4608	4358	3733
18"	450	9116	7808	8237	7055	7470	6398	5670	4856
20"	500	11590	9927	10440	8942	9820	8411	7499	6423
24"	600	17590	15066	16180	13858	14940	12796	10400	8908
28"	700	28000	23982	23400	20042	19082	16594	-	-
30"	750	31500	26980	30000	25695	23850	20428	-	-
32"	800	36000	30834	33300	28521	26715	22311	-	-
36"	900	45000	38543	42500	36401	32650	27965	-	-
40"	1000	56000	47964	51800	44367	-	-	-	-
42"	1050	61700	52846	53350	45694	-	-	-	-
48"	1200	81000	69377	73970	63355	-	-	-	-
56"	1400	110750	95840	-	-	-	-	-	-

Only reference data is provided in this catalogue. Specific model selection shall be based on actual working conditions. As products are continuously improved and updated, we shall have the right to change without notice.



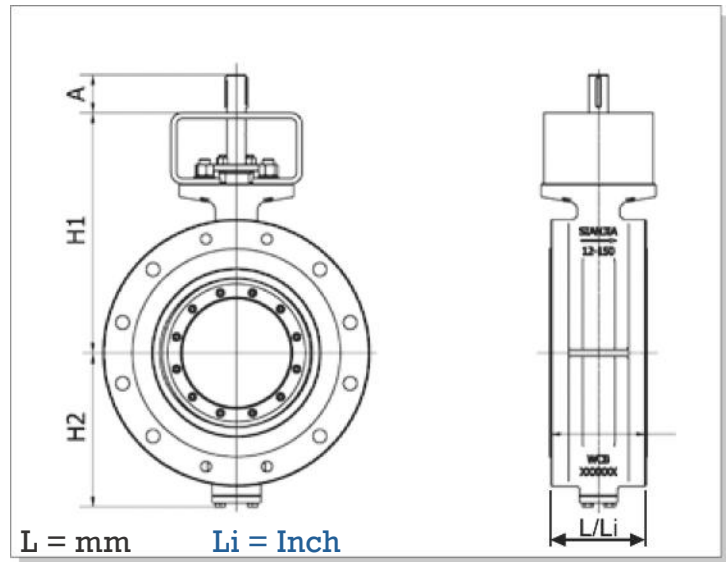
Triple Offset Butterfly Valve Z-TEK Series

Dimensions and Weights

Notes:

1. Recommended direction for valve assembly: The valve stem keeps horizontal or there is an angle of deflection with the vertical direction.
2. When the temperature of a medium is higher than 392°F (200°C), the valve shall keep warm so that the temperature difference between the valve body and internal parts is less than 212°F (100°C).
3. For any size and pressure rating not shown in the tables below, please consult Max-Seal for details.

Double Flanged Type



L = mm Li = Inch

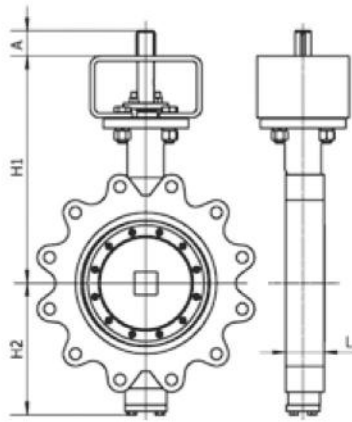
Size		Double Flange F-F ASME Cl. 150							Double Flange F-F ASME Cl. 300						
Inch	mm	H1	H2	A	L	Li	Weight kg	Weight lbs	H1	H2	A	L	Li	Weight kg	Weight lbs
3	80	260	125	36	114	4.48	22	10	260	125	36	114	4.48	22	10
4	100	290	141	36	127	5.00	28	13	290	141	36	127	5.00	33	15
6	150	330	165	40	140	5.51	40	18	350	189	40	140	5.51	61	28
8	200	385	214	40	152	5.98	61	28	415	236	56	152	5.98	86	39
10	250	425	243	40	165	6.49	83	38	455	272	56	165	6.49	100	45
12	300	490	286	56	178	7.00	127	58	490	310	80	178	7.00	175	79
14	350	525	312	56	190	7.48	166	75	545	335	80	190	7.48	284	129
16	400	454	350	80	216	8.50	198	90	575	389	140	216	8.50	340	154
18	450	580	383	80	222	8.74	229	104	660	422	140	222	8.74	487	221
20	500	580	414	80	229	9.01	311	141	700	462	140	229	9.01	529	240
24	600	690	484	140	267	10.51	433	196	785	531	160	267	10.51	834	378
28	700	795	556	140	292	11.49	702	318	880	617	160	292	11.49	1382	627
30	750	865	611	160	318	12.51	848	385	1000	671	180	318	12.51	1595	723
23	800	865	611	160	318	12.51	1015	460	1000	671	180	318	12.51	1805	819
36	900	1000	736	160	330	12.99	1495	678	1150	801	200	330	12.99	2215	1005
40	1000	1130	801	180	410	16.14	2035	923	1150	801	200	410	16.14	2270	1029
42	1050	1130	801	180	410	16.14	2155	977	1215	900	250	410	16.14	2756	1250
48	1200	1267	866	200	470	18.50	3045	1381	1300	991	280	470	18.50	3400	1542
56	1400	1595	1077	250	530	20.86	3280	1488	-	-	-	-	-	-	-

Size		Double Flange F-F ASME Cl. 600							Double Flange F-F ASME Cl. 900						
Inch	mm	H1	H2	A	L	Li	Weight kg	Weight lbs	H1	H2	A	L	Li	Weight kg	Weight lbs
3	80	234	136	40	180	7.08	32	15	-	-	-	-	-	-	-
4	100	298	185	40	190	7.48	66	30	-	-	-	-	-	-	-
6	150	355	238	56	210	8.26	121	55	335	238	56	225	8.85	167	76
8	200	405	263	80	230	9.05	198	90	466	293	80	275	10.82	297	135
10	250	485	328	140	250	9.84	298	135	520	345	140	325	12.79	385	175
12	300	538	365	140	270	10.62	378	171	620	378	160	375	14.76	588	267
14	350	610	390	140	290	11.41	445	202	683	456	160	425	16.73	795	361
16	400	680	440	160	310	12.20	670	304	730	492	160	475	18.70	1219	553
18	450	683	460	160	330	12.99	750	340	770	510	200	500	19.68	1441	654
20	500	795	526	200	350	13.77	1000	454	870	590	200	575	22.63	1924	873
24	600	885	623	200	390	15.35	1452	659	1030	673	200	675	26.57	2841	1288
28	700	1005	675	330	430	16.92	2020	916	-	-	-	-	-	-	-
32	800	1090	780	360	470	18.50	2960	1342	-	-	-	-	-	-	-
36	900	1240	860	410	510	20.07	3660	1660	-	-	-	-	-	-	-

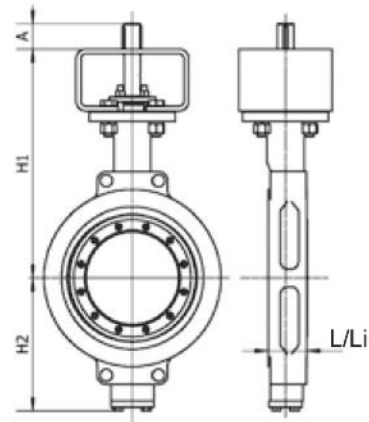


Triple Offset Butterfly Valve Z-TEK Series

Lug Type



Wafer Type



Size		Double Flange F-F ASME Cl. 150							Double Flange F-F ASME Cl. 300						
Inch	mm	H1	H2	A	L	Li	Weight kg	Weight lbs	H1	H2	A	L	Li	Weight kg	Weight lbs
3	80	260	125	36	48	1.88	13	6	260	125	36	48	1.88	18	8
4	100	290	141	36	54	2.12	17	8	290	141	36	54	2.12	22	10
6	150	330	165	40	57	2.24	23	10	350	189	40	59	2.32	41	19
8	200	385	214	40	64	2.51	36	16	415	236	56	73	2.87	56	25
10	250	425	243	40	71	2.79	49	22	455	272	56	83	3.26	77	35
12	300	490	286	56	81	3.18	83	38	490	310	80	92	3.62	119	54
14	350	525	312	56	92	3.62	117	53	545	335	80	117	4.60	254	115
16	400	545	350	80	102	4.01	160	73	575	389	140	133	5.23	300	136
18	450	580	383	80	114	4.48	194	88	660	422	140	149	5.86	455	206
20	500	580	414	80	127	5.00	270	122	700	462	140	159	6.25	499	226
24	600	690	484	140	154	6.06	387	176	785	531	160	181	7.12	788	357
28	700	795	556	140	165	6.49	614	278	-	-	-	-	-	-	-
30	750	865	611	160	190	7.48	678	307	-	-	-	-	-	-	-
32	800	865	611	160	190	7.48	801	363	-	-	-	-	-	-	-
36	900	1000	736	160	203	7.99	1015	460	-	-	-	-	-	-	-
40	1000	1130	801	180	216	8.50	1590	721	-	-	-	-	-	-	-
42	1050	1130	801	180	216	8.50	1802	817	-	-	-	-	-	-	-
48	1200	1267	866	200	254	10.00	2284	1036	-	-	-	-	-	-	-

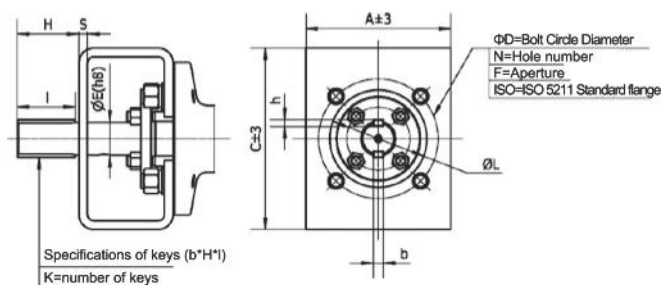
Size		Double Flange F-F ASME Cl. 600							Double Flange F-F ASME Cl. 900						
Inch	mm	H1	H2	A	L	Li	Weight kg	Weight lbs	H1	H2	A	L	Li	Weight kg	Weight lbs
3	80	260	125	36	48	1.88	14	6	260	125	36	48	1.88	14	6
4	100	290	141	36	54	2.12	15	7	290	141	36	54	2.12	15	7
6	150	330	165	40	57	2.24	20	9	350	189	40	59	2.32	29	13
8	200	385	214	40	64	2.51	34	15	415	236	56	73	2.87	50	23
10	250	425	243	40	71	2.79	45	20	455	272	56	83	3.26	75	34
12	300	490	286	56	81	3.18	73	33	490	310	80	92	3.62	109	49
14	350	525	312	56	92	3.62	97	44	454	335	80	117	4.60	164	74
16	400	545	350	80	102	4.01	123	56	475	389	140	133	5.23	228	103
18	450	580	383	80	114	4.48	164	74	660	422	140	149	5.86	285	129
20	500	580	414	80	127	5.00	220	100	700	462	140	159	6.25	343	156
24	600	690	484	140	154	6.06	324	147	785	531	160	181	7.12	513	233
28	700	795	556	140	165	6.49	537	244	-	-	-	-	-	-	-
30	750	865	611	160	190	7.48	594	269	-	-	-	-	-	-	-
32	800	865	611	160	190	7.48	650	295	-	-	-	-	-	-	-
36	900	1000	736	160	203	7.99	746	338	-	-	-	-	-	-	-
40	1000	1130	801	180	216	8.50	1172	532	-	-	-	-	-	-	-
42	1050	1130	801	180	216	8.50	1380	626	-	-	-	-	-	-	-
48	1200	1267	866	200	254	10.00	1860	844	-	-	-	-	-	-	-



Triple Offset Butterfly Valve Z-TEK Series

Top Works with ISO Bracket

All valves are Supplied with Actuator Mounting Brackets as Standard



ASME Class 150

Size		End Dimensions of Valve Stem				Dimensions of connecting Flange							
inch	mm	H	ΦE	K	b×h×l	ISO	A	C	S	ΦD	N	F	ΦL
3"	80	36	18	1	6×6×36	F10	120	140	6	102	4	12	72
4"	100	36	18	1	6×6×36	F10	120	140	6	102	4	12	72
6"	150	40	25	2	8×7×40	F14	175	175	8	140	4	18	103
8"	200	40	28	2	8×7×40	F14	175	175	8	140	4	18	103
10"	250	40	28	2	8×7×40	F14	175	175	8	140	4	18	103
12"	300	70	35	2	10×8×56	F16	200	240	10	165	4	22	133
14"	350	63	35	2	10×8×56	F16	200	250	12	165	4	22	133
16"	400	80	45	2	14×9×80	F16	200	250	12	165	4	22	133
18"	450	80	45	2	14×9×80	F16	200	250	12	165	4	22	133
20"	500	80	50	2	14×9×80	F25	300	340	15	254	8	18	204
24"	600	140	55	2	16×10×140	F25	300	340	15	254	8	18	204
28"	700	140	65	2	18×11×140	F25	300	400	20	254	8	18	204
30"	750	160	75	2	20×12×160	F30	350	410	20	298	8	22	234
32"	800	160	75	2	20×12×160	F30	350	410	20	298	8	22	234
36"	900	160	90	2	25×14×160	F35	415	550	30	356	8	32	263
40"	1000	180	110	2	28×16×180	F40	475	630	30	406	8	39	305
42"	1050	180	110	2	28×16×180	F40	475	630	30	406	8	39	305
48"	1200	200	120	2	32×18×200	F40	475	630	30	406	8	39	305
56"	1400	250	140	2	36×20×250	F48	600	700	35	483	12	39	375

ASME Class 300

Size		End Dimensions of Valve Stem				Dimensions of connecting Flange							
inch	mm	H	ΦE	K	b×h×l	ISO	A	C	S	ΦD	N	F	ΦL
3"	80	36	18	1	6×6×36	F10	120	140	6	102	4	12	72
4"	100	36	18	1	6×6×36	F10	120	140	6	102	4	12	72
6"	150	40	25	2	8×7×40	F14	175	175	8	140	4	18	103
8"	200	55	35	2	10×8×55	F16	200	240	10	165	4	22	133
10"	250	55	35	2	10×8×55	F16	200	250	12	165	4	22	133
12"	300	80	45	2	14×9×80	F16	200	260	15	165	4	22	133
14"	350	80	50	2	14×9×80	F25	300	340	15	254	8	18	204
16"	400	140	55	2	16×10×140	F25	300	340	15	254	8	18	204
18"	450	140	65	2	18×11×140	F30	350	400	20	298	8	22	234
20"	500	140	65	2	18×11×140	F30	350	400	20	298	8	22	234
24"	600	160	75	2	20×12×160	F30	350	410	20	298	8	22	234
28"	700	160	90	2	25×14×160	F35	415	550	30	356	8	32	263
32"	800	200	120	2	32×18×200	F40	475	630	30	406	8	39	305
36"	900	140	135	2	36×20×280	F40	475	630	30	406	8	39	305
40"	1000	280	160	2	40×22×280	F48	560	710	40	483	12	39	370
48"	1200	320	170	2	40×22×320	F60	700	800	40	603	20	39	470

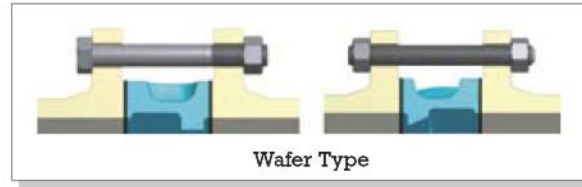
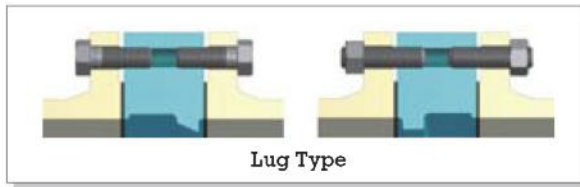
ASME Class 600

Size		End Dimensions of Valve Stem				Dimensions of connecting Flange							
inch	mm	H	ΦE	K	b×h×l	ISO	A	C	S	ΦD	N	F	ΦL
3"	80	40	20	1	6×6×40	F10	120	180	10	102	4	12	77
4"	100	40	30	2	8×7×40	F14	160	250	10	140	4	18	105
6"	150	65	40	2	12×8×63	F16	160	260	10	165	4	22	132
8"	200	80	45	2	14×9×80	F16	160	270	15	165	4	22	135
10"	250	110	55	2	16×10×110	F25	300	330	15	254	8	18	203
12"	300	140	60	2	18×11×140	F30	350	380	18	298	8	22	234
14"	350	140	70	2	20×12×140	F30	350	400	20	298	8	22	234
16"	400	160	75	2	20×12×160	F35	415	470	20	356	8	32	265
18"	450	160	90	2	25×14×160	F35	415	470	20	356	8	32	265
20"	500	200	100	2	28×16×200	F40	475	580	30	406	8	39	305
24"	600	220	120	2	32×18×220	F40	475	580	30	406	8	39	305
28"	700	330	160	2	40×22×320	F48	600	700	35	483	12	39	375
32"	800	360	180	2	45×25×350	F48	600	700	35	483	12	39	375
36"	900	410	200	2	45×25×400	F56	700	800	40	603	20	39	475



Triple Offset Butterfly Valve Z-TEK Series

Recommended Length of Flange Bolts



Size		Lug type F-F ASME CI.150					
		Hexagon Head Bolt			Stud		
Inch	mm	Quantity	Dimension	Length (inch)	Quantity	Dimension	Length (inch)
3"	80	4	5/8-11UNC	2	4	5/8-11UNC	2 1/4
		4	5/8-11UNC	1 1/2	4	5/8-11UNC	2 1/2
4"	100	8	5/8-11UNC	2	8	5/8-11UNC	2 1/4
		8	5/8-11UNC	1 3/4	8	5/8-11UNC	2 1/2
6"	150	8	3/4-10UNC	2 1/4	8	3/4-10UNC	3
		8	3/4-10UNC	2	8	3/4-10UNC	3 1/4
8"	200	8	3/4-10UNC	2 1/2	8	3/4-10UNC	3 1/2
		8	3/4-10UNC	2	8	3/4-10UNC	3
10"	300	12	7/8-9UNC	2 3/4	12	7/8-9UNC	4
		12	7/8-9UNC	2 1/4	12	7/8-9UNC	3 1/4
12"	350	12	7/8-9UNC	3	12	7/8-9UNC	4
		12	7/8-9UNC	2 1/2	12	7/8-9UNC	3 3/4
14"	400	12	1-BUN	3 1/2	12	1-BUN	4 3/4
		12	1-BUN	2 3/4	12	1-BUN	4
16"	450	16	1-BUN	3	16	1-BUN	4 1/4
		16	1-BUN	3 1/4	16	1-BUN	4
18"	500	16	1 1/8-BUN	3 3/4	16	1 1/8-BUN	5
		16	1 1/8-BUN	3 3/4	16	1 1/8-BUN	5
20"		16	1 1/8-BUN	4 1/4	16	1 1/8-BUN	6
		16	1 1/8-BUN	4	16	1 1/8-BUN	5 1/2
	600	4	1 1/8-BUN	3	4	1 1/8-BUN	5
		4	1 1/8-BUN	3	4	1 1/8-BUN	4 1/2
24"	750	20	1 1/8-BUN	5	20	1 1/8-BUN	6 3/4
		20	1 1/8-BUN	4 1/2	20	1 1/8-BUN	6
30"		24	1 1/8-BUN	7 1/2	24	1 1/8-BUN	7
		24	1 1/8-BUN	5 1/2	24	1 1/8-BUN	9
	900	4	1 1/8-BUN	4 1/4	4	1 1/8-BUN	5 1/2
		4	1 1/8-BUN	4 1/2	4	1 1/8-BUN	5
36"		28	1 1/2-BUN	7	28	1 1/2-BUN	9
		28	1 1/2-BUN	8	28	1 1/2-BUN	10
		4	1 1/2-BUN	5	4	1 1/2-BUN	7 1/2
		4	1 1/2-BUN	6	4	1 1/2-BUN	8

Size		Lug type F-F ASME CI.300					
		Hexagon Head Bolt			Stud		
Inch	mm	Quantity	Dimension	Length (inch)	Quantity	Dimension	Length (inch)
3"	80	8	3/4-10UNC	2 1/4	8	3/4-10UNC	3 1/4
		8	3/4-10UNC	1 3/4	8	3/4-10UNC	2 3/4
4"	100	8	3/4-10UNC	2 1/2	8	3/4-10UNC	3 1/4
		8	3/4-10UNC	2	8	3/4-10UNC	3
6"	150	12	3/4-10UNC	2 3/4	24	3/4-10UNC	3 1/2
		12	3/4-10UNC	2 1/4			
8"	200	12	7/8-9UNC	3 1/4	12	7/8-9UNC	4 1/2
		12	7/8-9UNC	2 1/2	12	7/8-9UNC	3 1/2
10"	250	16	1-BUN	3 1/4	16	1-BUN	5
		16	1-BUN	3	16	1-BUN	4 1/4
12"	300	16	1 1/8-BUN	4	16	1 1/8-BUN	5 1/2
		16	1 1/8-BUN	3 1/2	16	1 1/8-BUN	4 3/4
14"	350	16	1 1/8-BUN	4 1/2	16	1 1/8-BUN	6
		16	1 1/8-BUN	4	16	1 1/8-BUN	5 1/2
		4	1 1/8-BUN	3 1/4	4	1 1/8-BUN	4 3/4
		4	1 1/8-BUN	3	4	1 1/8-BUN	4 1/2
16"	400	16	1 1/8-BUN	5 1/2	16	1 1/8-BUN	6 1/2
		16	1 1/8-BUN	4	16	1 1/8-BUN	6 1/4
		4	1 1/8-BUN	3 1/2	4	1 1/8-BUN	5
		4	1 1/8-BUN	3	4	1 1/8-BUN	5
18"	450	40	1 1/8-BUN	5 1/2	40	1 1/8-BUN	6 3/4
		4	1 1/8-BUN	3 1/2	4	1 1/8-BUN	5
		4	1 1/8-BUN	3 1/2	4	1 1/8-BUN	5
20"	500	20	1 1/8-BUN	6	20	1 1/8-BUN	8
		20	1 1/8-BUN	4 1/2	20	1 1/8-BUN	6
		4	1 1/8-BUN	6	4	1 1/8-BUN	6
		4	1 1/8-BUN	4 1/2	4	1 1/8-BUN	7 3/4
24"	600	20	1 1/8-BUN	6 1/2	20	1 1/8-BUN	8
		20	1 1/8-BUN	6	20	1 1/8-BUN	8
		4	1 1/8-BUN	5	4	1 1/8-BUN	6 3/4
		4	1 1/8-BUN	4	4	1 1/8-BUN	6

Size		Wafer type F-F ASME CI.150					
		Hexagon Head Bolt			Stud		
Inch	mm	Quantity	Dimension	Length (inch)	Quantity	Dimension	Length (inch)
3"	80	4	5/8-11UNC	4 1/2	4	5/8-11UNC	5 1/4
4"	100	8	5/8-11UNC	4 3/4	8	5/8-11UNC	5 1/2
6"	150	8	3/4-10UNC	5 1/4	8	3/4-10UNC	6
8"	200	8	3/4-10UNC	5 3/4	8	3/4-10UNC	6 1/2
10"	250	12	7/8-9UNC	6 1/4	12	7/8-9UNC	7
12"	300	12	7/8-9UNC	7	12	7/8-9UNC	7 1/2
14"	350	12	1-BUN	7 1/2	12	1-BUN	8 1/2
16"	400	16	1-BUN	8	16	1-BUN	9
18"	450	16	1 1/8-BUN	9 1/4	16	1 1/8-BUN	10 1/2
20"	500	16	1 1/8-BUN	10	16	1 1/8-BUN	11
		4	1 1/8-BUN	3 1/2	4	1 1/8-BUN	5
		4	1 1/8-BUN	3	4	1 1/8-BUN	4 1/2
24"	600	20	1 1/8-BUN	11 1/2	20	1 1/8-BUN	12 1/2
30"	750	24	1 1/8-BUN	13 1/4	24	1 1/8-BUN	16
		4	1 1/8-BUN	3 1/2	4	1 1/8-BUN	5 1/4
		4	1 1/8-BUN	3 1/4	4	1 1/8-BUN	4 3/4
36"	900	28	1 1/2-BUN	15	28	1 1/2-BUN	18 1/4
		4	1 1/2-BUN	5	4	1 1/2-BUN	6 3/4
		4	1 1/2-BUN	4	4	1 1/2-BUN	6

Size		Wafer type F-F ASME CI.300					
		Hexagon Head Bolt			Stud		
Inch	mm	Quantity	Dimension	Length (inch)	Quantity	Dimension	Length (inch)
3"	80	8	3/4-10UNC	5	8	3/4-10UNC	5 3/4
4"	100	8	3/4-10UNC	5 1/2	8	3/4-10UNC	6 1/2
6"	150	12	3/4-10UNC	6	24	3/4-10UNC	7
8"	200	12	7/8-9UNC	7 1/4	12	7/8-9UNC	8
10"	250	12	1-BUN	8 1/4	16	1-BUN	9 1/2
		4	1-BUN	3 3/4	16	1-BUN	4 3/4
		4	1-BUN	3 1/4		1-BUN	4 1/4
12"	300	16	1 1/8-BUN	9	16	1 1/8-BUN	10
14"	350	16	1 1/8-BUN	10 1/4	16	1 1/8-BUN	11 1/2
		4	1 1/8-BUN	3 1/4	16	1 1/8-BUN	4 3/4
		4	1 1/8-BUN	3	4	1 1/8-BUN	4 1/2
16"	400	16	1 1/8-BUN	11 1/2	16	1 1/8-BUN	12 1/2
		4	1 1/8-BUN	3 1/4	16	1 1/8-BUN	4 3/4
		4	1 1/8-BUN	3	4	1 1/8-BUN	4 1/2
18"	450	20	1 1/8-BUN	12	40	1 1/8-BUN	13 1/2
		4	1 1/8-BUN	3 1/2	4	1 1/8-BUN	5
		4	1 1/8-BUN	3	4	1 1/8-BUN	4 3/4
20"	500	20	1 1/8-BUN	13	20	1 1/8-BUN	14
		8	1 1/8-BUN	4	20	1 1/8-BUN	5 1/2
24"	600	20	1 1/8-BUN	14 1/2	20	1 1/8-BUN	16
		4	1 1/8-BUN	4 3/4	20	1 1/8-BUN	6 1/2
		4	1 1/8-BUN	4 1/4	4	1 1/8-BUN	6 1/4



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